

Mathematical Analysis of Hybrid ESKF-TCN Framework

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1 Boundedness of Sensor Covariance

Lemma 1 (The Boundedness). *For a robust state estimation, the measurement covariance matrix R_{inv} must be positive definite and bounded:*

$$\Delta\mathcal{I}_{ext} = H^T R^{-1} H \geq \delta I > 0 \quad (1)$$

where δ is a strictly positive lower bound ensuring information gain.

1.1 TCN-based Uncertainty Modeling

The heteroscedastic uncertainty is predicted via a Temporal Convolutional Network (TCN) trained using the Negative Log-Likelihood (NLL) loss function. The diagonal covariance matrix $R_{TCN,k}$ is defined as:

$$R_{TCN,k} = \text{diag}(\sigma_{x,k}^2, \sigma_{y,k}^2, \sigma_{z,k}^2, \sigma_{qx,k}^2, \sigma_{qy,k}^2, \sigma_{qz,k}^2) \quad (2)$$

where each variance $\sigma_{i,k}^2$ is strictly bounded within the range $[\sigma_{min}^2, \sigma_{max}^2]$ to prevent numerical instability:

$$0 < \lambda_{min} \mathbf{I} \leq R_{min} \leq R_{TCN,k} \leq R_{max} < \infty \quad (3)$$

1.2 Robust Gating and Soft-ZUPT Mechanism

To handle statistical outliers, the system employs a Mahalanobis Gating and Hard Clamping mechanism. Furthermore, the zero-velocity update (ZUPT) is adaptively modulated via a Soft-ZUPT approach. The adaptive measurement covariance is dynamically updated by the zero-velocity probability $P_k \in [0, 1]$:

$$R_k = (1 - P_k)R_{high} + P_k R_{low} \quad (4)$$

This ensures that the information matrix $J_k = H^T R_{TCN,k}^{-1} H$ maintains its positive definiteness $\delta \mathbf{I} > 0$, guaranteeing the stability of the hybrid architecture.

2 Divergence of Pure Inertial Integration

2.1 Error State Dynamics

Let $\delta \mathbf{x} \in \mathbb{R}^n$ denote the error state vector, which includes errors in position, velocity, orientation, and sensor biases. In the absence of external corrections (pure integration), the discrete-time evolution of the error state is governed by:

$$\delta \mathbf{x}_{k+1} = \mathbf{F}_k \delta \mathbf{x}_k \quad (5)$$

where $\mathbf{F}_k$ represents the discrete-time state transition matrix derived from the system dynamics.

2.2 Information Decay and Rank Deficiency

The Fisher Information Matrix (FIM), denoted as $\mathcal{I}_k$, represents the total amount of information the system holds regarding its states. The recursive update of the FIM is given by:

$$\mathcal{I}_{k+1} = \mathbf{F}_k^{-T} \mathcal{I}_k \mathbf{F}_k^{-1} + \mathbf{H}_{k+1}^T \mathbf{R}_{k+1}^{-1} \mathbf{H}_{k+1} \quad (6)$$

In a pure inertial navigation scenario, there are no external measurements available, implying $\mathbf{H}_k = \mathbf{0}$. Consequently, the information update term vanishes:

$$\mathbf{H}_k^T \mathbf{R}_k^{-1} \mathbf{H}_k = \mathbf{0} \quad (7)$$

As a result, the FIM evolves purely based on system dynamics. Due to the inherent geometry of inertial sensors, states corresponding to the global position (δp) and yaw ($\delta \psi$) form the null space of the observation manifold. This leads to a rank deficiency in the information matrix:

$$\text{rank}(\mathcal{I}_k) < \dim(\delta \mathbf{x}) \quad (8)$$

2.3 Unbounded Drift Theorem

Since the error covariance matrix $\mathbf{P}_k$ is the inverse of the FIM ($\mathbf{P}_k = \mathcal{I}_k^{-1}$), the rank deficiency of $\mathcal{I}_k$ implies that the eigenvalues corresponding to the unobservable subspace diverge to infinity over time:

$$\lim_{t \rightarrow \infty} \lambda_{\max}(\mathbf{P}_k) = \infty \quad (9)$$

This formally proves that open-loop inertial integration inevitably leads to unbounded drift in unobservable states.

3 Reduction of CRLB via Hybrid Information Injection

3.1 Augmented Information Matrix

Let $\mathcal{I}_{base,k}$ denote the Fisher Information Matrix (FIM) of the baseline pure inertial navigation system. In the proposed hybrid architecture, the information matrix is augmented by external pseudo-measurements provided by the TCN-ESKF coupling:

$$\mathcal{I}_{hybrid,k} = \mathcal{I}_{base,k} + \Delta \mathcal{I}_{ext,k} \quad (10)$$

The external information gain, $\Delta \mathcal{I}_{ext,k}$, is derived from the pseudo-velocity observation equation:

$$\Delta \mathcal{I}_{ext,k} = \mathbf{H}^T \mathbf{R}_k^{-1} \mathbf{H} \quad (11)$$

3.2 Probabilistic Measurement Covariance

The measurement update logic is governed by a switching mechanism based on the zero-velocity probability $p_k \in [0, 1]$. The adaptive covariance $\mathbf{R}_k$ is defined as a convex combination of high and low uncertainty states:

$$\mathbf{R}_k = (1 - p_k) \mathbf{R}_{high} + p_k \mathbf{R}_{low} \quad (12)$$

Since $\mathbf{R}_{high}$ and $\mathbf{R}_{low}$ are symmetric positive definite (SPD) matrices, their convex combination $\mathbf{R}_k$ is strictly positive definite for all p_k . This ensures that the information gain $\Delta \mathcal{I}_{ext,k}$ is non-zero and bounded.

3.3 Cramér-Rao Lower Bound (CRLB) Reduction

The Cramér-Rao Lower Bound, which defines the theoretical minimum variance of any unbiased estimator, is the inverse of the FIM. By the Loewner order property of positive definite matrices ($A \succeq B \Rightarrow A^{-1} \preceq B^{-1}$), the addition of external information yields:

$$(\mathcal{I}_{base,k} + \Delta\mathcal{I}_{ext,k})^{-1} \preceq \mathcal{I}_{base,k}^{-1} \quad (13)$$

Consequently, the hybrid architecture strictly reduces the error lower bound:

$$CRLB_{hybrid} \leq CRLB_{base} \quad (14)$$

This probabilistic decay mechanism acts as a differentiable soft constraint, effectively minimizing the CRLB in the zero-velocity regime without introducing algorithmic discontinuities.

4 Stochastic Lyapunov Stability and UUB Analysis

4.1 Lyapunov Function Candidate

To prove the stochastic stability of the estimator, we define a Lyapunov function candidate V_k based on the weighted squared error form:

$$V_k(\delta\mathbf{x}_k) = \delta\mathbf{x}_k^T \mathbf{P}_k^{-1} \delta\mathbf{x}_k \quad (15)$$

where $\delta\mathbf{x}_k$ is the error state vector and $\mathbf{P}_k$ is the error covariance matrix. This scalar represents the "energy" or the magnitude of the estimation uncertainty in the system.

4.2 Energy Variation in Prediction Step

During the prediction step, process noise injects entropy into the system, inevitably increasing the uncertainty volume. The variation in V_k during prediction is bounded by the noise characteristics:

$$\Delta V_{pred} = V_{k|k-1} - V_{k-1|k-1} \leq \mu \quad (16)$$

where μ represents the maximum energy increase due to sensor noise and linearization errors.

4.3 Dissipative Energy Variation in Update Step

In the update step, the Ground Velocity Decay (GVD) mechanism and the TCN-based innovations apply a constructive mapping on the velocity error states. The network acts as a dissipative term in the energy function. Regardless of the switching state, the update step imposes a negative energy variation:

$$\Delta V_{update} = -\delta x_k^T (H^T R_k^{-1} H) \delta x_k \leq -\beta \|\delta x_k\|^2 \quad (17)$$

where $\beta = \lambda_{min}(H^T R_k^{-1} H) > 0$ is the minimum dissipation rate guaranteed by the strictly positive definite information matrix.

5 Stability Synthesis and Global Convergence

5.1 Total Energy Difference

Combining the effects of the Prediction and Update steps, the total energy variation of the Lyapunov function is given by:

$$\Delta V_{total} = \Delta V_{pred} + \Delta V_{update} \quad (18)$$

For the estimator to remain stable ($\Delta V_{total} < 0$), the dissipative term provided by the TCN-ESKF coupling must overpower the noise injection.

5.2 Uniform Ultimate Boundedness (UUB)

It follows that whenever the error magnitude exceeds a specific threshold defined by the noise levels μ and the dissipation rate β , the energy derivative becomes negative, forcing the error back toward the equilibrium. Consequently, the estimation error is **Uniformly Ultimately Bounded (UUB)**, ensuring that the trajectory drift remains within a strictly defined physical corridor.

6 Conclusion

6.1 Restatement of Premises

The theoretical validity of the Trajecto framework rests upon the following core premises:

1. **Bounded Measurement Noise:** Sensor uncertainties are strictly confined within stable numerical ranges.
2. **Statistical Consistency of TCN:** The neural network provides reliable, non-Markovian innovations that reflect real-world physical constraints.
3. **Hybrid Information Augmentation:** The continuous injection of external pseudo-measurements via Soft-ZUPT effectively counteracts the inherent rank deficiency of pure inertial systems.

6.2 Final Verdict

By combining the reduction of the Cramér-Rao Lower Bound (CRLB) with the proof of UUB stability, this work establishes that the proposed hybrid ESKF-TCN architecture is theoretically guaranteed to outperform pure inertial baselines. The system achieves a technical singularity where deep learning-driven innovations and deterministic physical laws converge to solve the long-standing problem of unbounded IMU drift.